

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, status LED, fault pull-ups

File: mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 4-wire (ADS1220)

File: measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: rs485.kicad_sch

DOTECH HMT500 – Oil Moisture Transmitter – Schematic v0.2

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-30 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), VDDA, CHASSIS. Inter-sheet signals: global labels.

VERIFY: TPS2660, LMR36006, PCAP04, DAC8760, TPS26611 pin numbers are placeholders (datasheets not accessible when drawn).

TBD footprints: M Connect 8P, CMC, GDT, feedthrough, buck inductor, the five VERIFY ICs.

VERIFY=YES parts: placeholder pin numbers (TPS2660, LMR36006, PCAP04, DAC8760, TPS26611)

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

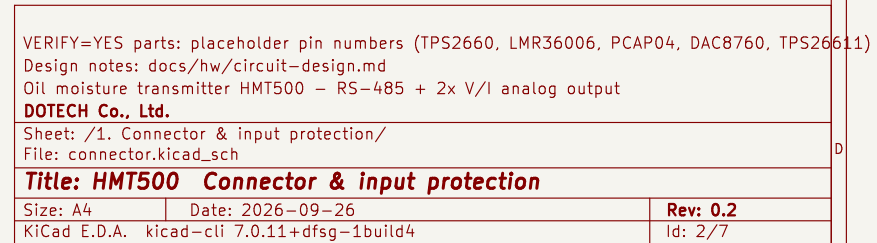
Sheet: /

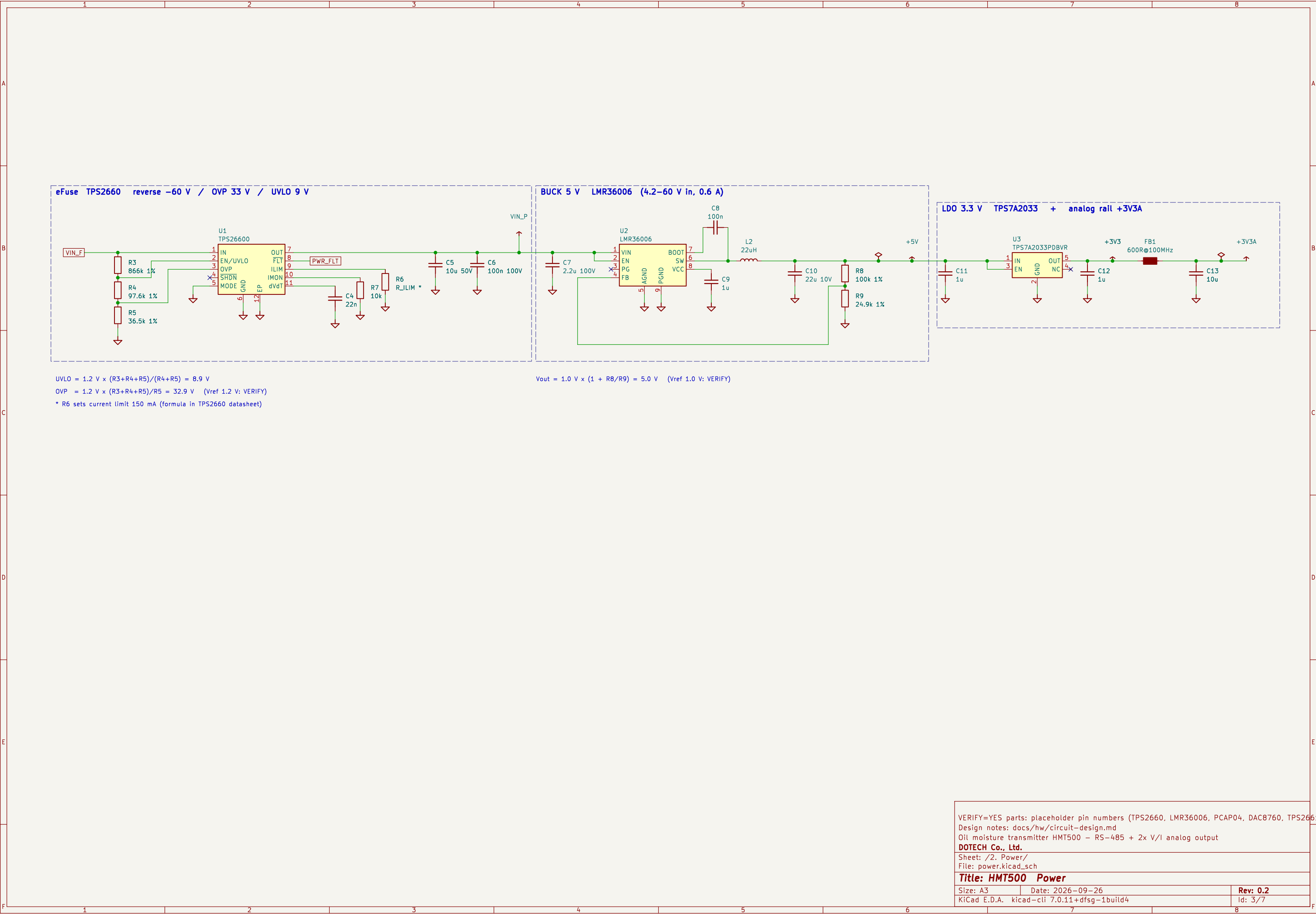
File: HMT500.kicad_sch

Title: HMT500 Top

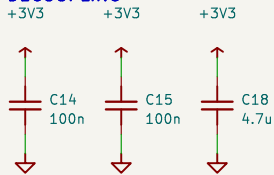
Size: A3Date: 2026-09-26Rev: 0.2

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4Id: 1/7



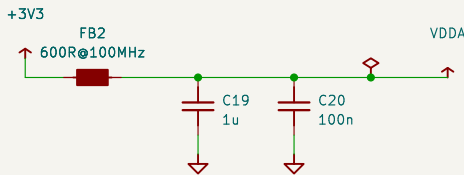


MCU DECOUPLING



100 nF at VDD and VBAT, 4.7 uF bulk

VDDA FILTER



MCU NOTES (STM32G0B1)

BOOT0 shares PA14/SWCLK – no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

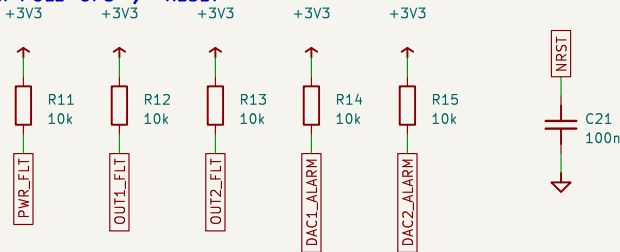
-> boot from flash (RS-485 bootloader 0x08000000).

VDDA bonded to VDD in LQFP48; VREF+ via FB2.

Cortex-M0+: no SWO (J2 pin 6 NC).

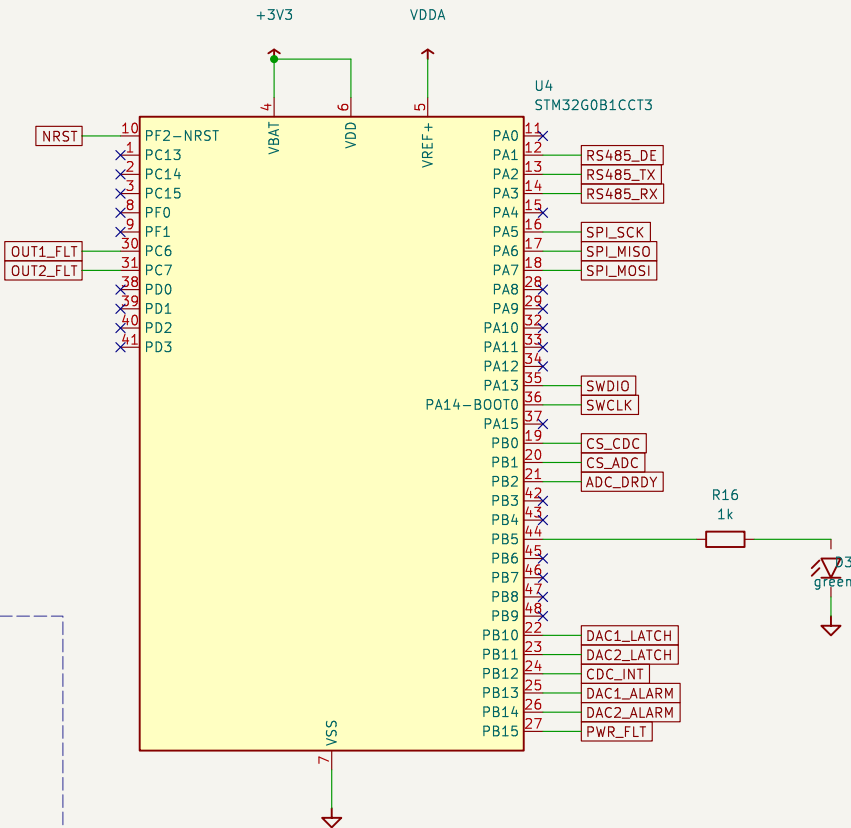
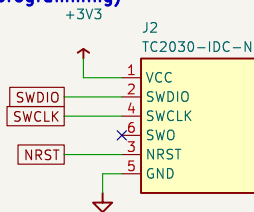
Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

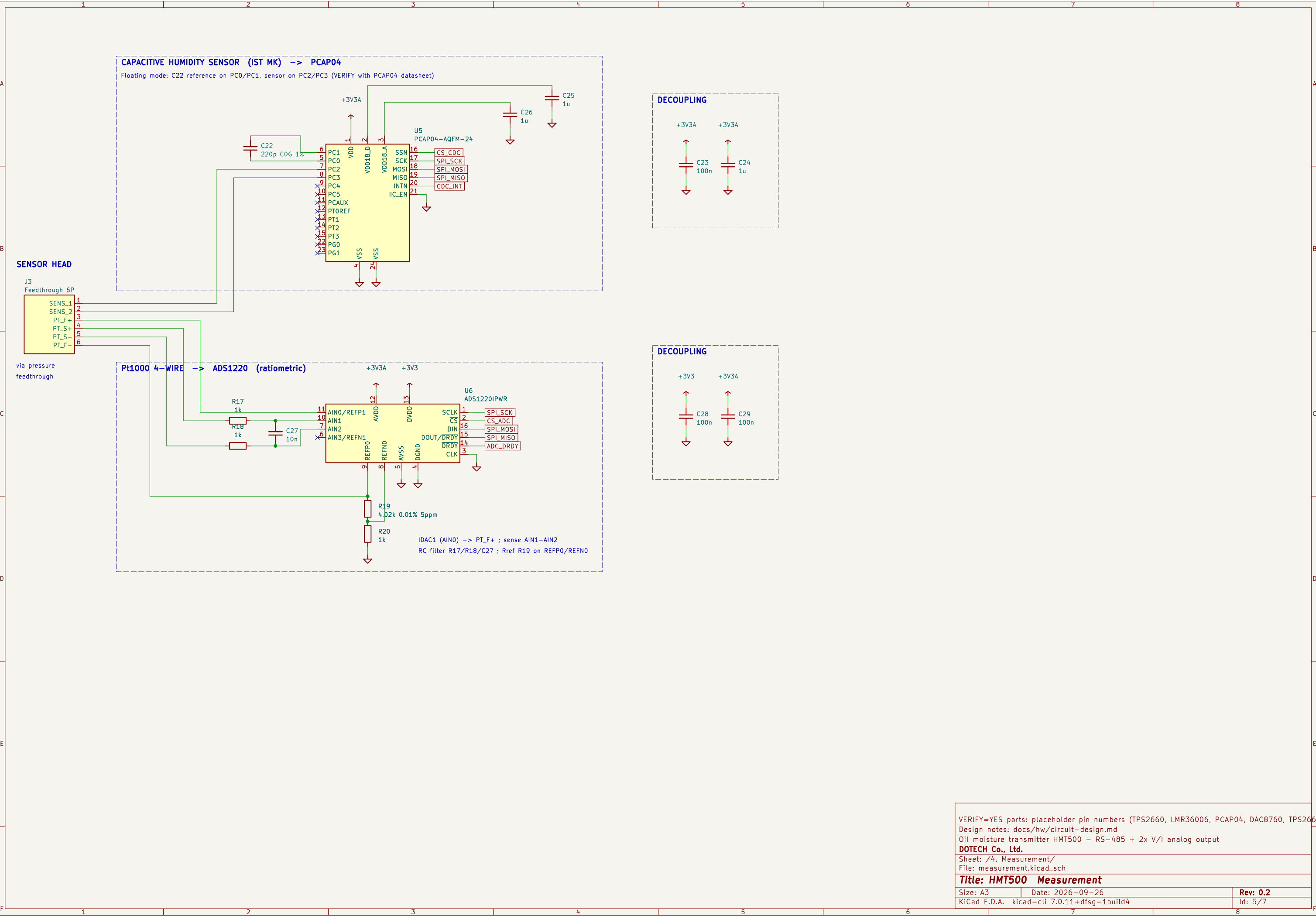
FAULT PULL-UPS / RESET

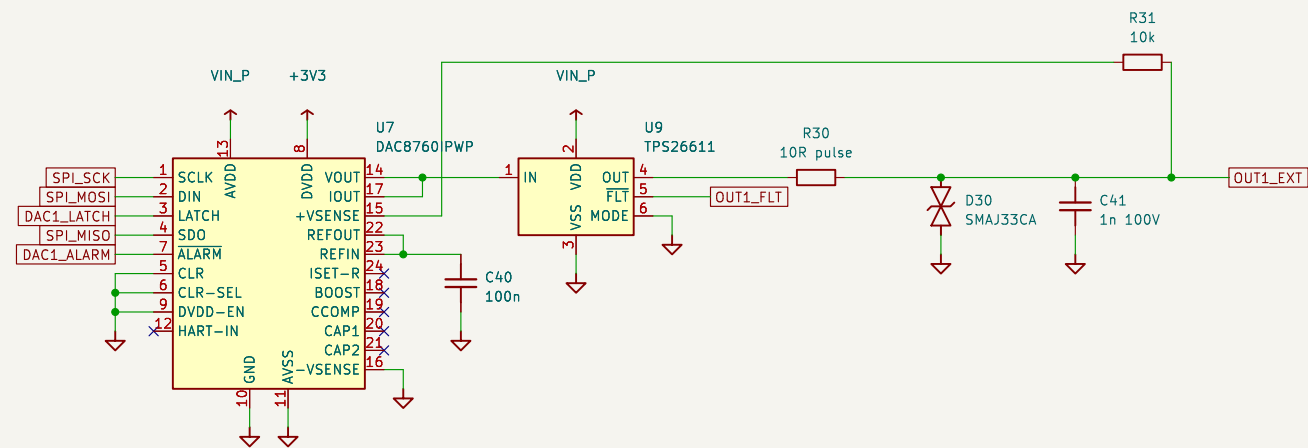


Open-drain fault lines pulled up to +3V3

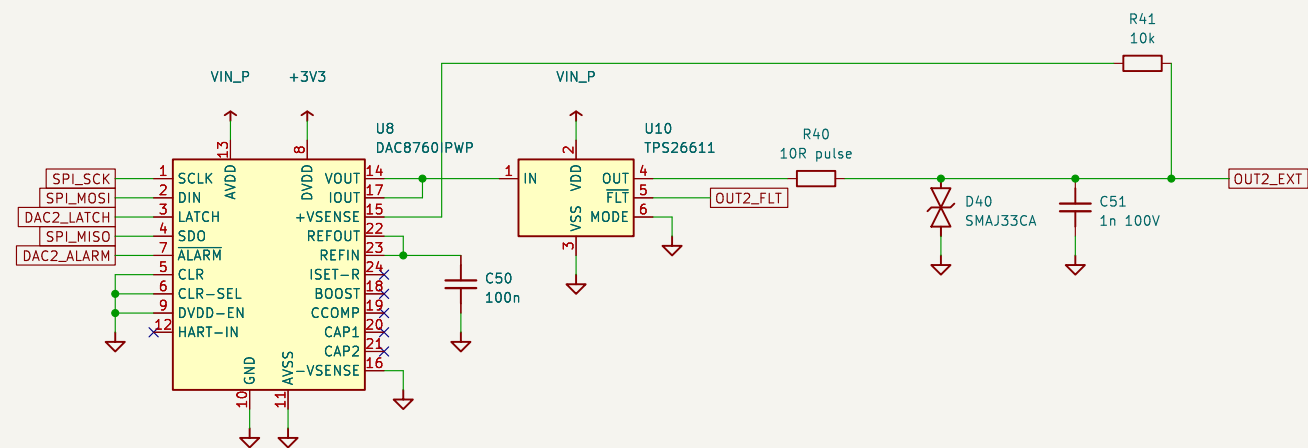
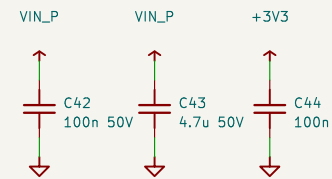
SWD (production programming)



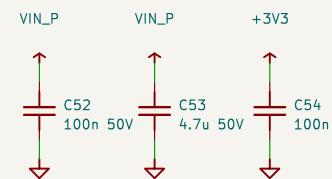




VOUT and IOUT combined (per TI TIDUBK2 – VERIFY). +VSENSE taken after R30 to cancel series drop.



VOUT and IOUT combined (per TI TIDUBK2 – VERIFY). +VSENSE taken after R40 to cancel series drop.

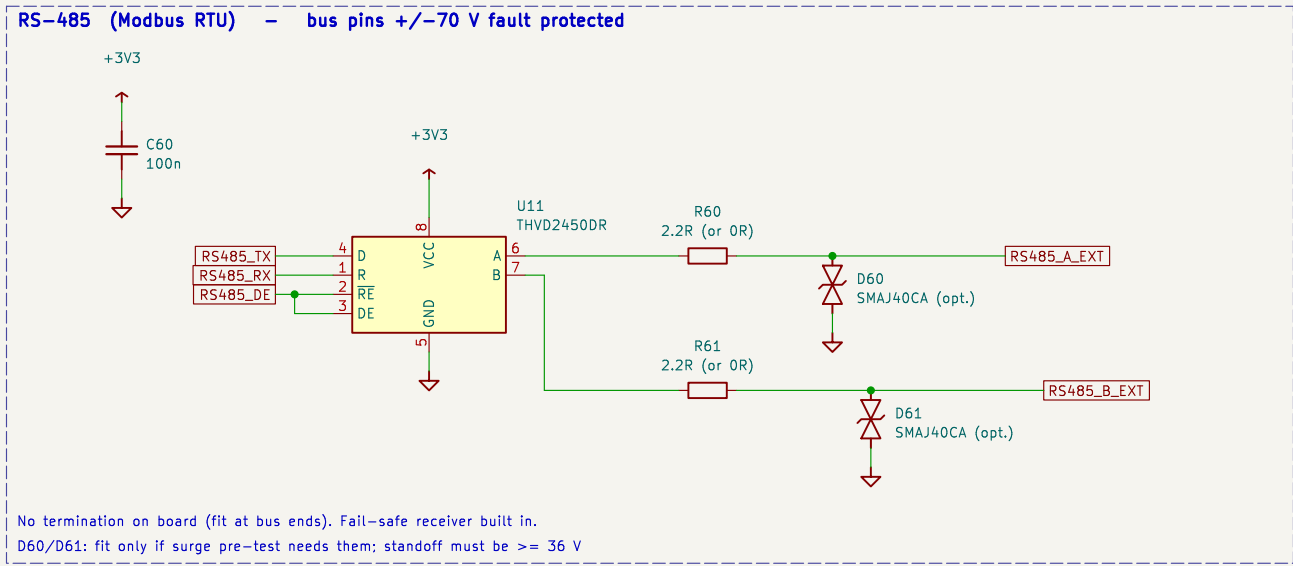


VERIFY=YES parts: placeholder pin numbers (TPS2660, LMR36006, PCAP04, DAC8760, TPS26611)
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output
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Sheet: /5. Analog outputs/
File: analog_out.kicad_sch

Title: HMT500 Analog outputs

Size: A3	Date: 2026-09-26
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	

Rev: 0.2
Id: 6/7



VERIFY=YES parts: placeholder pin numbers (TPS2660, LMR36006, PCAP04, DAC8760, TPS26611)

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /6. RS-485/

File: rs485.kicad_sch

Title: HMT500 RS-485

Size: A4

Date: 2026-09-26

Rev: 0.2

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 7/7